SUSTAINABLE LUXURY

Quiet, high-craft living that makes its own energy and water and closes the food + waste loop.

20.0 kWp
Solar PV + 30 kWh battery

~77%
Electricity from the sun

80 m³
Rainwater storage for the garden

~75%
Dry-season garden water from rain

Well
Primary house water; mains backup only

4-6 Hens
Eggs + compost loop

MOVES, STATUS + ROUGH COST (THB)

MOVE	STATUS	ROUGH THB	IMPACT
Rooftop solar 20 kWp + 30 kWh battery (E-102)	Included	1,020,000	~77% of electricity, ~13 t CO2/yr avoided
Rainwater cisterns 80 m³ + smart drip (P-102)	Included	710,000	~75% of dry-season garden water
Drilled well + treatment, mains as backup only (P-103)	Included	561,000	Water self-sufficiency
Home controller: solar-timed pool/irrigation, leak shut-off (E-103)	Included	347,000	10-15% energy, leak protection
Heat-pump water heater 300 L	Included	85,000	~3x less energy than resistive
Passive cooling: deep eaves, courtyard cross-ventilation, ceiling fans, light roofs	Included	90,000	Cuts AC hours 20-30%
ERV fresh air with enthalpy core	Included	180,000	Healthy air at low cooling cost
Greywater reuse to subsurface drip (GW-1)	Included	60,000	~150-250 L/day reused
Kitchen garden, raised beds, compost + 4-6 hens	Included	160,000	Food + closes the waste loop
Indoor hydroponic herb nook (solar-powered LEDs)	Included	589,000	Year-round herbs, zero food miles
EV-ready: 7.4 kW charger + spare conduit to carport	Included	45,000	Solar-charged driving
Low-carbon materials: AAC block, slag/fly-ash cement blend, plantation teak, charred timber (no paint), lime plaster	Included	Cost-neutral	~15-25% less embodied carbon (cost-neutral)
Bidirectional EV charger (V2H backup)	Optional	350,000	Car becomes a 60 kWh home battery
Solar-thermal / heat-pump pool heating on surplus	Optional	120,000	Warm pool without grid power
Green (sedum) roof on studio, + acoustic + cooling	Optional	250,000	Cooler studio, habitat
Biogas digester for kitchen + coop waste	Optional	40,000	Cooking gas, fertilizer
Earth-tube pre-cooling for ERV intake	Optional	150,000	2-4 °C cooler fresh air

Included total about 3,847,000 THB · Optional add-ons about 910,000 THB (concept-level, 2026 prices, excl. VAT)

Sheet references: E-102 Solar, E-103 Controls, P-102 Rainwater + Kitchen Garden, P-103 Well + Treatment, M-101 HVAC. In the 3D viewer, the Solar, Water + Rainwater, Controls and Landscape + Farm layers show each system in place.

WEBERMAN RESIDENCE

Mountain courtyard house, Thailand
Owner: Max Weberman

ISSUE
Design Development

NOT FOR CONSTRUCTION

DATE
11 Oct 2026

SCALE
NTS

MODEL
bim.json (from plan.json)

UNITS
Feet + inches (mm for members)

REVISION
DD-01

NOTES
Costs are rough order-of-magnitude for budgeting; confirm with Thai contractors.
Carbon figures use the Thai grid factor ~0.5 kg CO2/kWh.

SHEET TITLE
Sustainability Summary

SHEET
G-002